University of Tennessee College of Veterinary Medicine

Department of Small Animal Clinical Science
Veterinary Physical Rehabilitation

INITIAL EVALUATION Date: ________ Chief Complaint: ___________________________
Weight: _________
Diagnosis: ____ Surgery Date: _______ Attending / RDVM: ______________________________
Affected Limb: RF LF RR LR N/A

SUBJECTIVE AND HISTORY

General appearance and disposition:
____
____
____

Onset of problem/injury:
____
____

Past pertinent medical history:
____
____
____

Home environment/Baseline activity level:
____
____

Diet/medications/supplements:
____
____

Additional Information and Owner goals:
____
____

OBJECTIVE FINDINGS

PT ORTHOPEDIC LAMENESS (POSTURE/STANCE)

0=normal stance/no lameness
1=slightly abnormal stance (PWB)
2=moderately abnormal stance (PWB-TTWB)
3=severely abnormal stance (TTWB-NWB)
4=unable/unwilling to bear weight (NWB)

PT ORTHOPEDIC LAMENESS (AT WALK & TROT)

0=Normal walk/trot, no lameness
1=Slight or intermittent lameness
2=Obvious weight bearing lameness
3=Severe weight bearing lameness
4=Intermittent non-weight bearing lameness
5=Continuous non-weight bearing lameness

PT NEUROLOGIC GRADING SCHEME

5=normal strength/coordination no obvious dysfunction
4=can stand to support minimal paraparesis and ataxia
3=can stand to support but freq. stumbles and falls; mild paraparesis and ataxia
2=unable to stand or support; with assistance moves limbs readily but freq. stumbles/mod ataxia and paresis
1=unable to stand or support; slight movement when supported by the tail; severe paresis
0=absence of purposeful movement; tetra- or paraplegic

Rehab Orthopedic Measures

Lameness stance (0-4; 0=normal): Pre-op_____ Post-op _____ Current _____
Lameness walk (0-5; 0=normal): Pre-op_____ Post-op _____ Current _____
Lameness trot (0-5; 0=normal):Pre-op_____ Post-op _____ Current _____

Figure 12-2 Rehabilitation evaluation form. Courtesy University of Tennessee College of Veterinary Medicine.

Continued

Rehab Neurologic Measures

Neurologic Grading Scheme (0-5; 0=normal): Pre-op_____ Post-op _____ Current _____

Gait

____

____

Standing limb position:

____

Sitting limb position:

____

Willingness to raise contralateral limb: YES NO RELUCTANT UNABLE

Force Plate: YES NO Analysis Results: ______________________________________

Skin/Incision: ____

Discoloration: _______________ Temperature to touch: _____________Other: ____________

Swelling/Edema/Muscle Mass

Circumference of limb: Affected ____cm Unaffected ___cm Site & Position: ______________

Comments:____

Additional measurements (if taken):

____

____

Range of Motion

Joint (normal range)	R Flex/Ext	L Flex/Ext	End Feel/Comments	Test Position
Carpus (32°-196°)	____ /____	____/____	_________________	_________________
Elbow (36°-166°)	____/____	____/____	_________________	_________________
Shoulder (57°-165°)	____/____	____/____	_________________	_________________
Tarsus (38°-165°)	____/____	____/____	_________________	_________________
Stifle (41°-162°)	____/____	____/____	_________________	_________________
Hip (50°-162°)	____/____	____/____	_________________	_________________

Pain/Comments:____

Functional Assessment

____

____

Palpation

____

____

Special Tests and Results

____

____

Figure 12-2, cont'd

ASSESSMENT

____

____

____

PLAN/ RECOMMENDATIONS

____

____

____

____

____

Short-term goals (time frame)

____

____

____

Long-term goals (time frame)

____

____

____

____

Evaluator Signature/Date

____

Figure 12-2, cont'd

Figure 12-3 A 6-month-old Labrador retriever is recovering from extracapsular stabilization of his left stifle joint after avulsion of his cranial cruciate ligament. The dog is undergoing static weight shifting exercises (*A*), walking in an underwater treadmill (*B*), trotting on a treadmill (*C*), and walking with an elastic band eliminating the external rotation present in his right pelvic limb during the recovery period (*D*).

assessments are regularly performed and documented. Measurements, such as joint range of motion, muscle girth, and lameness assessments at a walk, trot, or gallop, are frequently assessed and related to changes in function. A functional questionnaire also may be used at the beginning, during, and end of a course of rehabilitation. In all cases, close communication is maintained between the rehabilitation practitioners and referring veterinarian regarding the dog's treatment plan, changes to treatment plan, and overall progress.

Delivery of Care

The owner, trainer, therapist, therapist assistant, and others may deliver aspects of the physical rehabilitation plan of care. Some tasks may be delegated to the owner. Other more technical tasks should be performed by the therapist. Specifically owners may administer some exercises to their dogs and may apply cold packs. Some owners may have the ability to perform tasks such as neuromuscular electrical stimulation or other treatments. The care must be supervised and coordinated by the clinicians who assess the patient, decide who will perform the treatments, and determine how the treatment plan should be modified over time, including increases or decreases in exercise duration and frequency, and initiating or discontinuing modalities. Feedback (i.e., the severity of current clinical signs and the presence or absence of new clinical signs) must be collected from all involved in the dog's care, and adjustments in the patient care plan must be communicated to all involved.

REFERENCES

1. Guide to physical therapist practice, ed 2, *Phys Ther* 81:9-744, 2003.
2. Shelbourne KD, Nitz P: Accelerated rehabilitation after anterior cruciate ligament reconstruction, *Am J Sport Med* 18:292-299, 1990.

3. Sherrington C, Lord SR: Home exercise to improve strength and walking velocity after hip fracture: a randomized controlled trial, *Arch Phys Med Rehabil* 78:208-212, 1997.
4. Moffet H, Collet JP, Shapiro SH et al: Effectiveness of intensive rehabilitation on functional ability and quality of life after first total knee arthroplasty: a single blind randomized controlled study, *Arch Phys Med Rehabil* 85:546-556, 2004.
5. Ostelo RWJG, de Vet HCW, Waddell G et al: *Rehabilitation after lumbar disc surgery (Cochrane review). The Cochrane Library, issue 2*, Oxford, UK, 2002, Update Software.
6. Yeung EW, Yeung SS: A systematic review of interventions to prevent lower limb soft tissue running injuries, *Br J Sport Med* 35:383-389, 2001.
7. Chu KS, Eng JJ, Dawson AS, Harris JE et al: Water-based exercise for cardiovascular fitness in people with chronic stroke: a randomized controlled trial, *Arch Phys Med Rehabil* 85:870-874, 2004.
8. Dias RC, Dias JM, Ramos LR: Impact of an exercise and walking protocol on the quality of life for elderly people with OA of the knee, *Physiother Res Int* 8:121-130, 2003.
9. Munin MC, Rudy TE, Glynn NW et al: Early inpatient rehabilitation after elective hip and knee arthroplasty, *J Am Med Assoc* 279:847-852, 1998.
10. Aure OF, Hoel Nilsen J, Vasseljen O: Manual therapy and exercise therapy in patients with chronic low back pain: a randomized, controlled trial with 1-year follow up, *Spine* 28:525-532, 2003.
11. Ntoumenopoulos G, Presneill JJ, McElholum M et al: Chest physiotherapy for the prevention of ventilator-associated pneumonia, *Int Care Med* 28:850-856, 2002.
12. Di Fabio RP, Choi T, Soderberg J et al: Health-related quality of life for patient with progressive multiple sclerosis: influence of rehabilitation, *Phys Ther* 77:1704-1716, 1997.
13. Baker LL, Chambers TR, DeMuth SK et al: Effects of electrical stimulation on wound healing in patients with diabetic ulcers, *Diabetes Care* 20:405-412, 1997.
14. Engsberg JR, Ross SA, Park TS: Changes in ankle spasticity and strength following selective dorsal rhizotomy and physical therapy for spastic cerebral palsy, *J Neurosurg* 91:727-732, 1991.
15. Brosseau L, Yonge KA, Robinson V et al: *Thermotherapy for treatment of osteoarthritis*. Cochrane Database of Systematic Reviews 2003, Issue 4. Art. No.: CD004522. doi: 10.1002/14651858.CD004522.
16. Cyriax JH: Textbook of orthopaedic medicine, vol 1: Diagnosis of soft tissue lesions, ed 8, London, 1982, Ballière Tindall.

13 Assessing and Measuring Outcomes

Darryl L. Millis and David Levine

Assessing the outcome of treatments, including physical rehabilitation, is essential to determine how an animal is progressing and to assess the effectiveness of treatment protocols. Review of data and evaluation of protocols are necessary so changes can be initiated to improve outcomes. Assessments should consist of objective data whenever possible because owners and veterinarians often believe a patient is doing better than the data suggest. In addition, documentation of progress is important to provide incentive for owners to continue rehabilitation and justify continued treatment.

Several measurements are useful for assessing outcomes, including the ability to perform functional activities of daily living, gait analysis, pedometers, joint function, muscle mass and strength, body composition, pain assessment, impressions of owners and veterinarians, return to function, and quality of life.

Activities of Daily Living

Activities of daily living (ADLs) can be defined as the activities the animal must perform independently on a daily basis. Despite the best treatments, rehabilitation facilities, and personnel, some conditions are so serious that complete recovery cannot be expected. In these situations, the owner must be informed of a reasonable expected outcome. The rehabilitation goals should be realistic and should concentrate on performing basic life functions, such as eating and drinking with no or minimal assistance, changing body positions without assistance, rising from a sitting position, and walking outside to urinate and defecate with no or minimal assistance. In some cases even these goals may not be met. Progress may be slow but it should be recorded and rehabilitation plans should concentrate on activities to help achieve these rehabilitation goals.

The expected ADLs of working animals undergoing rehabilitation, such as guard dogs, police dogs, and assistance dogs for the disabled, should be discussed and clearly understood at the initial evaluation. A dog that is expected to work for 8 consecutive hours may need intensive rehabilitation often progressing up to several hours/day to ready it to return to work.

Measurements of ADLs are sometimes difficult. The timed get up and go (TUG) test is a measure of mobility that has been studied and used extensively in human rehabilitation.[1] The person is asked to stand up from a standard chair and walk a distance of 3 meters, turn around, and walk back to the chair and sit down again. This test involves a number of tasks, such as standing from a seating position, walking, turning, stopping, and sitting down, which are all important tasks needed for a person to be independently mobile. Normal time required to finish the test is between 7 and 10 seconds. Similar tests could be developed for dogs, but because the TUG depends on a person's mental ability and understanding of what is expected, it must be modified for dogs. (Dogs competing in some sporting events such as fly-ball may be an exception because of their training and understanding of what is expected.) Nevertheless a dog can be timed during a test of rising from a sitting position, walking on leash for a distance, turning, and then sitting. For the average house pet other objective activities that could be measured include how far a dog can walk or trot on a leash before needing to sit or lay down. These can all be compared over time to assess function and mobility.

Gait Analysis

Weight Bearing at a Stance

Evaluation of a dog's stance gives information regarding willingness to place complete weight on an affected limb. Many dogs may have no visible or only very mild lameness at a walk or trot, but will not bear equal weight on the limbs when standing. Weight bearing at a stance may be assessed by observing the placement of a foot in relation to the contralateral front or rear foot. In severe cases the dog may hold the foot completely or partially off the ground. More commonly the dog places a moderate amount of weight on the foot but not complete weight. The toes may point out and when the limb is gently pushed forward it may be moved more easily than the contralateral limb when it is gently pushed. Alternatively the evaluator may place both hands under the fore or rear feet with the palm facing the pads, and the relative amount of weight bearing may be assessed. Weight bearing at a stance may be incorporated into global lameness scores.

An inexpensive method of acquiring more quantitative information regarding weight bearing at a stance is to have

Figure 13-1 Measure of static weight-bearing forces using a computerized stance analyzer (Petsafe, Knoxville, TN).

Box 13-1 Lameness Evaluation at a Walk and a Trot

Lameness Evaluation at a Walk

0 Walks normally
1 Slight lameness
2 Obvious weight-bearing lameness
3 Severe weight-bearing lameness
4 Intermittent non–weight-bearing lameness
5 Continuous non–weight-bearing lameness

Lameness Evaluation at a Trot

0 Trots normally
1 Slight lameness
2 Obvious weight-bearing lameness
3 Severe weight-bearing lameness
4 Intermittent non–weight-bearing lameness
5 Continuous non–weight-bearing lameness

a dog stand with each limb on common household scales. It is important to be certain that the dog is standing squarely in a standard position, with each limb on a scale. Dogs should also undergo a period of acclimation to the scales so that data collection is valid. Scales should be calibrated periodically to standard weights to be certain that accurate weights are recorded. One study found that bathroom scales were a reliable, simple, and cost-effective objective method for measuring static weight bearing in dogs with osteoarthritis in the hindlimbs.[2]

Computerized devices are available that simultaneously measure static weight bearing of all four limbs (Figure 13-1). As with any test, proper data collection is crucial to obtaining reliable information. One study of various conditions during data collection concluded that to achieve consistent results, one standard method of conditions during data collection should be used.[3] Factors to consider include the position of the handler, proximity to walls, and local environment. We recommend having the dog in a quiet environment with minimal noise or distractions, with the dog facing either the center of the room or directly toward, but some distance away from, a wall. Any other walls should be at least 3 to 6 feet from the dog. It is critical that the dog stand squarely and symmetrically, with each foot in the middle of pressure pad. In most cases, lifting the dog's back or front limbs off the ground temporarily and then placing them back on the ground will result in square limb placement. The head and neck should be relaxed in a neutral position and facing straight ahead. We have found that dogs respond well to having the handler facing forward and straddling the dog over its back. Another option is for the handler to squat or kneel directly in front of the dog. The handler's hands can then help to guide the dog's head in a straight position, being careful to avoid touching the dog; the hands merely act as blinders to help keep the head in a straight position. Following these guidelines results in very consistent data collection and has very good correlation with ground reaction forces measured with a force plate.[4,5]

Lameness Scores

Lameness scores are a vital clinical outcome assessment tool. The walk and trot are the gaits most commonly evaluated because the speed of the limbs during movement is easier to assess without specialized equipment, and these gaits are symmetric, making identification of a lame limb easier. The reader is referred to the chapter on orthopedic and neurologic evaluation for information regarding identification of lameness. It is important to separate the walk from the trot when scoring lameness. Dogs are generally less lame at a walk because less force is placed on the limb at this less strenuous gait. Trotting may accentuate a lameness that is mild to moderate at a walk because of the greater forces placed on the limb with increased velocity. Having separate lameness scores for the walk and trot allows finer discrimination of gait analysis and may be more sensitive for detecting subtle improvements during rehabilitation (Box 13-1).

Although lameness scales are simple, commonly used, and inexpensive, they may not be accurate in evaluating subtle or mild lameness. One study compared a numerical scale and visual analog scale (VAS) with ground reaction forces and concluded that subjective scoring scales do not replace force plate gait analysis in dogs with surgically induced lameness.[6] Agreement of experienced evaluators with ground reaction forces was low unless lameness was severe. In addition each observer had his or her own individually unique scale. Therefore observers must stay the same during the evaluation of a patient for accurate analyses. Another study of induced hind limb lameness of dogs found that subjective evaluation of lameness varied greatly between different observers, and there was poor agreement with objective ground reaction force measurements.[7] Similar results regarding estimating the severity of lameness using a VAS have been found in dogs with fragmented coronoid processes.[8] However, owners may not be able to assign a valid VAS score.[9] Therefore, this form of evaluation is better performed by people experienced in evaluating lameness and pain.

An alternative approach may be to use a questionnaire to assess pain and lameness in dogs. One study evaluated a variety of questions using a VAS and found significant correlation with peak vertical force and vertical impulse.[10] Responses to questions recorded on a standard 10-cm VAS that had significant correlation with ground reaction forces included:

- Overall owner assessment for preceding month
- Mood of dog during preceding month
- Attitude of dog during preceding month
- Frequency of postures of a happy dog
- Change in amount of daily activity
- Willingness to play voluntarily
- How often dog gets exercise
- Stiffness when arising for the day
- Stiffness at end of the day (after activities)
- Indicates lameness when walking
- Pain when turning suddenly while walking

The Helsinki Chronic Pain Index also found mood, play, locomotion while walking and trotting, getting up, difficulty moving after rest, and difficulty moving after major activity to be valuable in helping to identify dysfunction with osteoarthritis.[11] Vocalization, galloping, jumping, and laying down are other activities that help identify dogs with osteoarthritis.

A Canine Brief Pain Index (CBPI) has also been developed to evaluate chronic pain in dogs with osteoarthritis, and consists of 11 questions related to pain, function, and overall impression of quality of life; a 10-point scale is used to indicate the severity of the condition for each of the 11 questions.[12] While the CBPI is able to detect improvement in pain scores of dogs with osteoarthritis after treatment with an NSAID,[13] there is not significant correlation of CBPI scores with ground reaction forces as measured with a force platform, suggesting that owners evaluate things other than lameness in assessing improvement with treatment.[14]

In summary, although lameness scores are commonly used and clinically useful in some circumstances, care must be used in interpreting the scores. In general, agreement among individuals making assessments is greatest in dogs with severe lameness and no lameness. There is generally poor agreement in mild cases of lameness, and there is poor correlation of lameness scores with objective measurement of ground reaction forces using a force platform. VAS scores may offer some advantages, but experience in lameness evaluation is important in assigning values. Finally, various pain indexes may be useful in assessing treatment effects in dogs with osteoarthritis. However, there is relatively poor correlation with ground reaction forces, likely because a variety of functional items other than just lameness are assessed by owners when using indexes.

Figure 13-2 Measurement of ground reaction forces using a force plate.

Kinetic (Force Plate) Analysis of Gait

Kinetic evaluation of gait involves the measurement of ground reaction forces with a force plate or platform (Figure 13-2). It is an objective, repeatable measure of weight bearing on limbs when proper technique is used for data collection. Lameness can be compared over a period of time without relying on memory of previous assessments because data may be stored on a computer. Although measurement of ground reaction forces is a reliable and well-accepted method of determining the degree of weight bearing on the limbs, it is an artificial situation, and some dogs may display different clinical signs in a home environment. Force platform systems are available at many veterinary colleges and some private practices. It is important that appropriate software be used for quadruped animals.

Force plates use either strain gauges or piezoelectric crystals. The force plate is either mounted on a platform or embedded in the floor so that it is even with the surface of the floor. A runway of adequate length is essential. Most systems have timer lights that are triggered as the handler and dog approach and cross the force plate to allow the calculation of mean velocity and acceleration. Control of velocity and acceleration within appropriate parameters is essential for repeatable data collection, because these greatly affect the force placed on each limb. The force plate is connected to a computer that calculates ground reaction forces. Force plates may also be embedded in treadmills so that consecutive footfalls can be recorded, with the gait being very rhythmic. However, some systems may not be able to accurately evaluate braking and propulsion forces.

Pressure pads or mats are also used for evaluation of canine gait. Many commercially available systems have an adequate number of sensors to detect ground reaction forces with reasonable accuracy. One advantage is that they can measure data from several consecutive footfalls. Another advantage is the ability to determine stride length, cadence, and average velocity.

The most useful forces measured are the peak vertical force (Z_{Peak}) and vertical impulse ($Z_{Impulse}$) (Figure 13-3). Other forces that may be useful are the peak braking ($Y_{A\ Peak}$) and propulsion ($Y_{B\ Peak}$) forces and braking ($Y_{A\ Impulse}$) and propulsion ($Y_{B\ Impulse}$) impulses (Figure 13-4). Medial-lateral forces (X_{Peak}) and impulse ($X_{Impulse}$) are likely too small and variable to be clinically useful. Forces may be measured during stance, walking, or trotting.

It is essential that appropriate technique be used during data collection. In addition to consistent velocity and acceleration targets, an experienced handler is necessary to minimize intertrial variability, although there is little or no difference in measured forces between experienced handlers. The handler should be between the dog's head and shoulder, and the dog should be gaited without undue tension or pulling on the leash. The dog should not alter the gait, throw its head, turn the head, lunge, or make other sudden movements as it approaches and crosses the force plate or pressure mat. The head carriage should be in a neutral position to avoid shifting weight to the front or rear limbs.

In general, dogs bear approximately 30% of their body weight on each front limb and 20% on each rear limb while in a standing position with the limbs placed squarely under the body. Walking at a velocity of 0.7 to 1.0 m/sec results in forces equivalent to approximately 60% of body weight on each forelimb and 40% on each rear limb in a medium to large dog. Increasing the velocity to a trot of 1.7 to 2.0 m/sec results in weight bearing of 100% to 120% of body weight on each forelimb and 65% to 70% of body weight on each hindlimb in a similar size dog. Reduced weight bearing on an individual limb may result in mild weight shifts to the other limbs.

Kinematic (Motion) Analysis of Gait

Kinematic or motion analysis of gait is a powerful tool that can be used to measure flexion and extension angles of

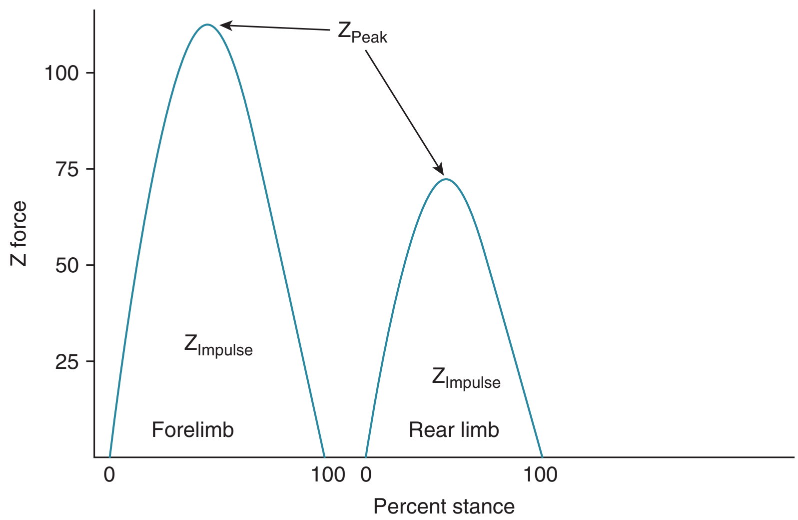


Figure 13-3 Graphic depiction of peak vertical force (Z_{Peak}) and vertical impulse ($Z_{Impulse}$).

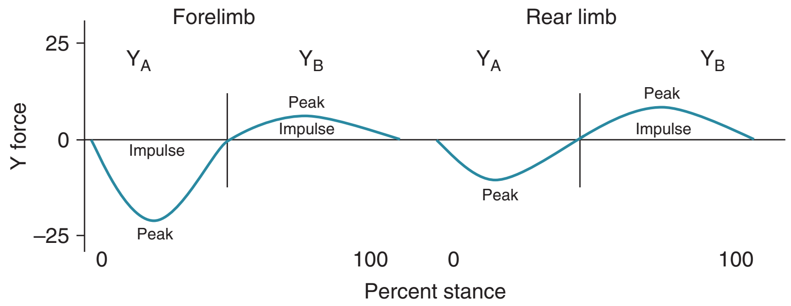


Figure 13-4 Graphic depiction of peak braking ($Y_{A\ Peak}$) and propulsion ($Y_{B\ Peak}$) forces and braking ($Y_{A\ Impulse}$) and propulsion ($Y_{B\ Impulse}$) impulses.

Figure 13-5 The dog is instrumented for kinematic assessment of joint motion and stride characteristics.

joints during gait, stride length, and other parameters of stride. It is usually combined with kinetic gait analysis. There is limited availability of three-dimensional kinematic gait analysis because the necessary equipment is sophisticated and expensive. Two-dimensional systems are less expensive, but they do not capture rotation and circumduction of limbs during gait, although simple flexion and extension of joints may be captured.

Attention to detail is critical for kinematic gait analysis. Most systems use a number of reflective devices that are attached to the dogs at very specific anatomic points to allow repeatable data collection at different times (Figure 13-5). Motion of the markers in relation to the joints is determined with a series of cameras interfaced with a computer. Software for quadruped animals allows reconstruction of a walking "stick figure" on the computer screen. The software is then used to calculate a number of measurements, including flexion and extension angles of joints during gait, angular velocity and acceleration of joints, and stride length and frequency.

Noninvasive, computer-assisted, three-dimensional kinematic gait analysis has been used to study lameness associated with cranial cruciate ligament rupture in dogs.[15] Dynamic flexion and extension angles and angular velocities were calculated for the hip, stifle, and hock joints. Distance and temporal variables were also determined. Mean flexion and extension curves were developed for all joints, and the changes in movement that occurred over time after cruciate rupture were compared. Each joint had a characteristic pattern of flexion and extension movement that changed with cruciate rupture. The stifle joint angle was more flexed throughout stance and early swing phase of stride and failed to extend in late stance. Angular velocity of the stifle joint was damped throughout stance phase, with extension velocity almost negligible. The hip and hock joint angles, in contrast with the stifle joint angle, were extended more during stance phase. Stride length and frequency also varied significantly after cruciate rupture. A change in the pattern of joint movement appeared to occur in which the hip and hock joints compensated for the dysfunction of the stifle joint.

Knowledge of kinematic parameters may allow more effective rehabilitation of patients following surgery or those with chronic conditions. In particular, changes that are noted during the course of treatment may allow for stepping up the level of activity, or more important, may signal changes that occur in other joints as a result of fatigue or overuse. Knowledge of these data may help prevent injuries to other joints or limbs. Recently, studies have been conducted to evaluate joint motion during therapeutic exercises.[16-21] The kinematics of various therapeutic exercises are discussed in the biomechanics of rehabilitation chapter.

Kinetic and kinematic gait analysis techniques have been combined using an inverse dynamics technique to estimate net joint moments and power. This technique has been used to evaluate tibial plateau leveling osteotomy and a lateral fabella-tibial technique to stabilize stifle joints in a cranial cruciate ligament transection model,[22] Labrador retrievers with and without cranial cruciate ligament disease,[23] and forelimb joints.[24] This form of gait evaluation is time-consuming and requires specialized software, but it can provide valuable information.

Pedometers

Most outcome assessment measurements are performed in a clinic in which equipment and personnel are located. However, in many cases, the animal temporarily becomes more active in the clinical environment as a result of traveling, exposure to other animals, and being in a strange environment. In these situations the dog often appears to be better than has been reported in the home environment. A pedometer may give some indication of a dog's activity level in its home environment. Most pedometers have a pendulum that swings back and forth with the normal gait cycle and indicates the number of steps taken. In our experience pedometers may be up to 85% to 90% accurate with regard to counting the number of steps taken by a dog. One study confirmed that pedometers are reasonably accurate, with the exceptions that at a walk, pedometers overestimated the number of steps by approximately 17% in large and medium dogs and underestimated actual number of steps by approximately 7% in small dogs.[25] Although these researchers found that no significant differences between pedometer-recorded and actual number of steps were detected when dogs trotted or ran, our experience has been that the step count is artificially increased when dogs run or climb stairs.

Proper placement of the pedometer is crucial. We have found that placing the pedometer just proximal to the elbow most accurately indicates the number of steps taken

Figure 13-6 Proper placement of a pedometer is crucial. Placing the pedometer just proximal to the elbow is one method that has been used to count the number of steps taken.

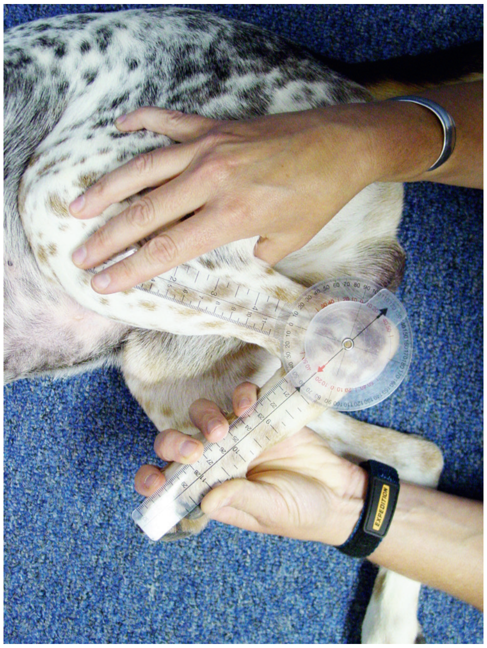


Figure 13-7 Joint angles of flexion and extension are measured using a goniometer.

(Figure 13-6). In the study by Chan et al., however, pedometers were worn around the dog's neck by means of an adjustable lightweight chain.[25] However, we have found that when dogs shake their head or scratch their neck, the step count on the pedometer is artificially increased. Because pedometers may not be not accurate in all dogs, it is recommended that the dog be walked 100 steps; the number of steps indicated by the pedometer is compared to see if the pedometer is reasonably accurate to record a dog's activity level. An additional test to determine accuracy is to place the pedometer and then allow the dog to have free activity while 100 steps are counted. Because the pedometer may double-count when the dog is trotting, running, or going up and down stairs, it is used only as a semiquantitative indicator of home activity. Environmental conditions are another major factor in the activity of dogs. Dogs that spend some time outdoors may not be as active if it is raining or cold, or the weather is otherwise inclement. Therefore comparative assessments should be made on days when the weather is as similar as possible.

Recently, accelerometers have been used to estimate activity in dogs. Because of the differences in technology between pedometers and accelerometers, placement at various sites on dogs can give acceptable results. One study found that placement on a collar in the ventral neck region gives acceptable results for home activity monitoring.[26] In addition, accelerometers were able to distinguish the intensity of activity in dogs.[27] Another group found that the use of accelerometers was able to detect an increase of 20% in home activity after treatment for osteoarthritis, while placebo-treated animals had no increase in activity.[28] Activity and location may also be monitored relatively inexpensively using GPS technology for pets.[29]

Joint Function

Joint motion may be evaluated using both objective and subjective assessments. The primary motion of a joint is the movement of bones as a whole, such as occurs with stifle flexion, and is termed physiologic or osteokinematic motion. The quantity of joint flexion and extension motion is measured using a goniometer (Figure 13-7). Unlike human range-of-motion measurements, the actual geometric angles are measured in dogs because the many different body types and conformation in individual dogs make estimation of the neutral or "0-degree" position difficult. Measuring the actual angle eliminates the need to estimate the "normal" standing angle; therefore there are no negative angles.

The maximum angles of extension and flexion are those angles of greatest joint excursion. Normal angles of maximum flexion and extension have been reported for the Labrador retriever and cat (see Appendix 1).[30,31] Excellent inter-tester and intra-tester reliability was reported in those studies. In addition, measurements made in dogs and cats had very good correlation with measurements made from radiographs.[30,31]

Measuring maximum angles may involve some discomfort. An animal experiencing discomfort is unlikely to use the limb at those angles while ambulating. Therefore measuring the comfortable range of motion may be more clinically applicable. To measure the comfortable range of motion, the joint is slowly flexed until the first indication of discomfort, such as tensing the muscles, pulling the limb away, or turning the head slightly, is noted. The joint is then slowly extended until the first indication of discomfort is noted. These angles are recorded. The mean of three independent measurements is used to be certain that the measured angles are reproducible. Physical rehabilitation may influence the return of joint motion following surgery, such as cranial cruciate ligament stabilization surgery, and joint motion is associated with function; therefore, measuring joint motion is important.

The quality of joint motion is more subjective and involves the assessment of joint biomechanics, crepitus, and pain during motion. The more subtle motions occurring at the surface of the joints are termed accessory or arthrokinematic motions. Examples of these motions are glide (slide), roll, spin, distraction or traction, and compression or approximation. Glides are shear or sliding motions of opposing articular surfaces. A normal amount of glide occurs in normal functioning joints. Glides at joint surfaces often are imposed interventions using joint mobilization techniques to regain normal motion in a joint with pathology. Joint surface geometry, soft tissue resistance, and external forces all affect glide. Rolls involve one bone rolling on another, such as the femoral condyles rolling on the tibial plateau. Gliding motion in combination with rolling is needed for normal joint motion. Spins are joint surface motions that result in continual contact of a single area of articular cartilage on adjacent articular cartilage within a joint. Distraction or traction accessory motions are tensile (pulling apart) movements between bones. Compressive or approximation accessory motions are compressive (pushing together) movements between bones. The quality of the end feel during joint motion may indicate abnormalities such as restriction by fibrous tissue, excess joint capsule, bone, or cartilage. Crepitus is often associated with surface irregularities in articular cartilage or periarticular changes, such as occur in osteoarthritis. The sensations of crepitus palpated during joint motion have been described as being similar to crinkling a piece of cellophane. Other sensations such as cracking, snapping, or popping may indicate abnormalities such as a torn meniscus.

Joint Laxity

Assessment of joint laxity in dogs is generally limited to qualitative evaluation, although some quantitative evaluations have been performed. In general joint laxity may occur as a result of developmental conditions (e.g., hip dysplasia), trauma, or pathologic degeneration of ligaments such as the cranial cruciate ligament.

Damage to collateral supporting structures is most easily assessed by placing varus and valgus stresses on the affected joint. The joints most commonly affected with collateral ligament damage include the hock, stifle, carpus, elbow, and shoulder. The digits are occasionally involved. Assessment is most accurate when the joint is placed in full extension to prevent inadvertent internal or external rotation of the joint that may be mistaken for varus or valgus movement. Breed and age differences exist regarding normal varus and valgus motion, especially in young animals, so comparison with the contralateral normal limb is helpful.

Hip joint laxity is most commonly assessed in young, growing dogs as an indication of early hip dysplasia. Subjective assessment is commonly performed using the Ortolani maneuver to create subluxation of the hip joint. A more quantitative method of assessing hip joint laxity is the use of PennHIP, in which controlled pressure is applied to the hip joint to create subluxation. A radiograph is made with the coxofemoral joint in this position, and measurements are made to quantify the degree of laxity, known as the hip distraction index. The hip distraction index has correlation with the development of hip dysplasia in some breeds of dogs.

Stifle joint laxity is clinically assessed by palpating for cranial drawer motion. There should be no drawer motion in normal dogs, with the exception of puppies, which have some drawer motion that comes to an abrupt stop at the end of drawer. Methods to quantify drawer motion, which are not commonly used in dogs, include measurement of direct cranial drawer, quantitative cranial tibial thrust, and use of instrumented devices.[32] Direct cranial drawer is determined by simply marking the position of the tibial tuberosity with the stifle in reduction on a piece of graph paper, and then remarking the position of the tibial tuberosity with full cranial drawer. Caution must be used to prevent motion of the femur and to keep the marking device perpendicular to the graph paper. Quantitative cranial tibial thrust is measured in a similar fashion, except that the stifle is kept in a fixed position (usually standing angle) while the hock is flexed. This results in cranial displacement of the tibial tuberosity if the cranial cruciate ligament is ruptured. Marks are made on graph paper with the tibia in a reduced position and with the tibia in full cranial tibial thrust. Care is taken to keep the caudal aspect of the femur in contact with a solid surface so the only motion that occurs is the cranial displacement of the tibia.

Instrumented devices for knee-drawer tests have been used in humans to measure shifts of the tibial tuberosity relative to the patella.[33-35] The total anterior and posterior displacement produced by anterior and posterior loads of 20 lb is measured from a reference position. Studies have evaluated the effects of variables on the accuracy and reproducibility of anterior-posterior drawer measurements. Reproducibility is principally affected by deviations in subject positioning. In addition to drawer motion, inadvertent knee flexion and tibial rotation may occur and result in incorrect measurements, despite attempts to prevent these. The effects of different observers, time sequences, different days, and muscle relaxation have been studied and contribute to error rates of 5% to 15%. Stress radiography, a relatively sensitive method of measuring drawer motion, has been used to compare instrumented systems in humans with tears of the anterior cruciate ligament (ACL). In one study stress radiography was superior to instrumented arthrometer testing for determining cruciate ligament status.[36] Similar studies have not been performed in dogs, but instrumented knee-drawer devices are difficult to use in dogs and require practice to obtain reproducible results. Pediatric devices may be easier to use because the smaller size may allow a better fit of the instrument to the dog's limb.

Muscle Mass and Strength

Assessment of muscle is important in physical rehabilitation. Muscle mass, muscle strength, and muscle injury may be assessed to help evaluate the patient's progress. Regaining muscle mass and strength following injury is important to help improve function and prevent further injury to joints and other soft tissues. The degree of muscle atrophy is also a reasonable indication of limb use. Therefore careful attention to muscle assessment provides a great deal of information regarding the progress of the rehabilitation patient.

Muscle Mass

Muscle mass measurements are a useful outcome measure in veterinary rehabilitation. Muscle mass indicates limb use and is associated with muscle strength. Muscle mass may be estimated with limb circumference measurements, ultrasound, computed tomography (CT), magnetic resonance imaging (MRI), and dual-energy x-ray absorptiometry (DEXA).

Measurement of limb circumference is an indirect method of assessing changes in muscle mass and is inexpensive, quick, and easily performed on clinical patients. Acceptable results depend on using standard, repeatable methods of measuring limb circumference. A measuring tape with a spring tension device is useful in measuring limb circumference to improve consistent tension on the tape when making measurements (Figure 13-8).

One study evaluated the effect of limb position, clipping hair, sedation, and different evaluators on thigh circumference measurements at two different locations before and after stifle stabilization surgery.[37] Thigh length was determined by measuring from the tip of the greater trochanter to the distal aspect of the lateral fabella. Circumference was determined at points equal to 50% and 70% of thigh length, measuring from the greater trochanter distally. Measurements were technically easier when made at the 70% location because the skin of the flank did not impede measurements (Figure 13-9). Clipping had little effect on thigh circumference measurements, but dogs with short hair coats were used (Figure 13-10). Measurements may be affected to a greater extent in dogs with long hair.

Figure 13-8 A measuring tape with a spring tension device is used to measure limb circumference. Use of this instrument allows the consistent placement of tension on the tape when making measurements.

Limb position is important. Full flexion of the stifle results in greater thigh circumference measurements as compared with measurements made with the limb placed at a functional standing angle or with the stifle fully extended (Figure 13-11). Extension of the joint causes the muscle bellies to elongate, whereas flexion results in shortening and bunching of muscle bellies, creating greater thigh circumference. Measurements made in awake dogs with the stifle extended are not significantly different from those made following heavy sedation, as long as the dogs are not overly tense, probably because the muscles are in an elongated position when the stifle is extended (Figure 13-12). Thigh circumference measurements are sensitive enough to detect muscle atrophy 2 weeks after stifle surgery (Figure 13-13). Similar measurements may be obtained by independent evaluators as long as standard technique is used and the evaluators practice before obtaining actual measurements.

The association of limb circumference with actual muscle mass is important if it is to be useful in the evaluation of rehabilitation patients. Human male cadavers were subjected to comprehensive anthropometry and weighing of all skeletal muscle.[38] Limb circumference had good correlation with total skeletal muscle mass for the forearm (r = .96), mid-thigh (r = .94), calf (r = .84), and midarm (r = .82).

Thigh circumference also has significant correlation with actual muscle mass in dogs (Figure 13-14). Even greater correlation may be obtained when thigh length is

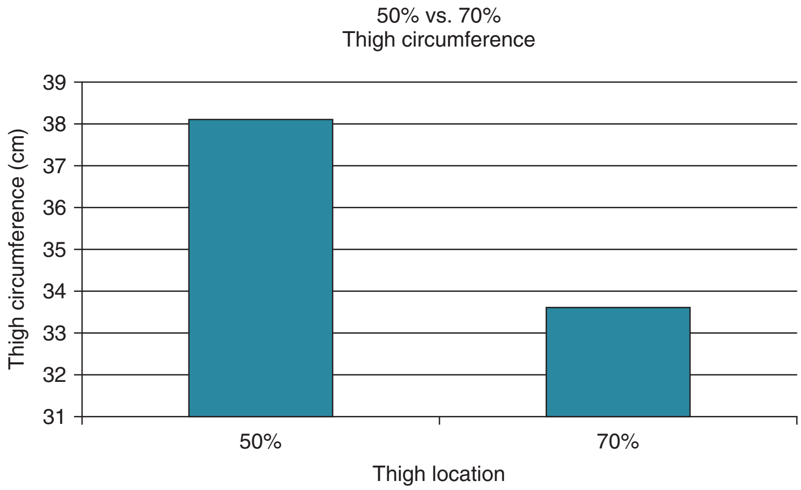


Figure 13-9 Thigh circumference measurements at the 50% and 70% thigh length. Although the 50% leg has greater thigh circumference because of greater muscle mass in that location, the measurements are technically more difficult at that location in some dogs because of the presence of the skin of the flank.

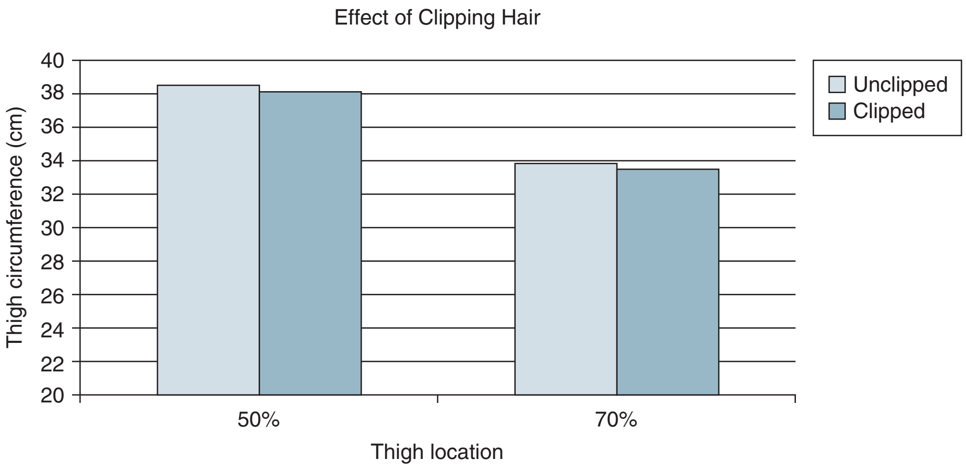


Figure 13-10 Clipping the hair has little effect on thigh circumference measurements in dogs with short hair coats. The effect of clipping would likely be greater in dogs with long hair.

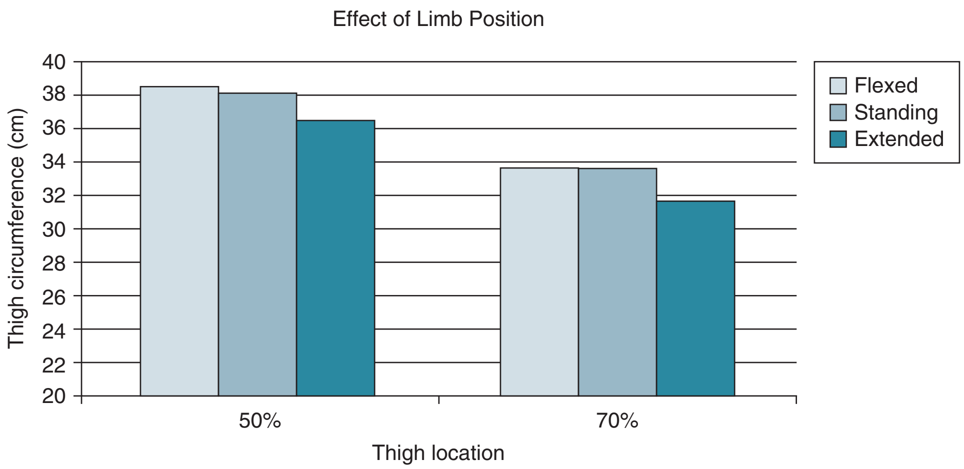


Figure 13-11 Thigh circumference is less with extension of the stifle because the muscles are elongated. Flexion of the limb results in greater thigh circumference, but the measurements are more difficult because of shortening of the hamstring muscles and difficulty in applying the tape measure.

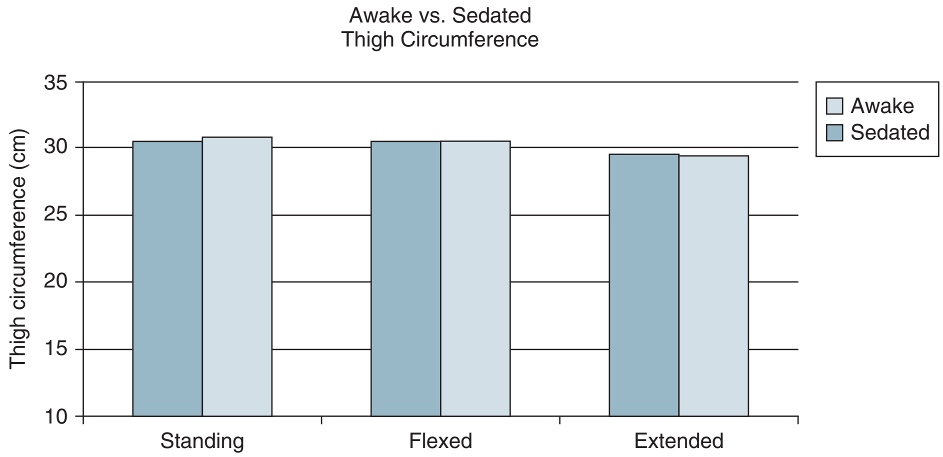


Figure 13-12 Measurements made in awake dogs with the stifle extended are not significantly different from those made following heavy sedation if the patient is not very tense, probably because the muscles are in an elongated position when the stifle is extended.

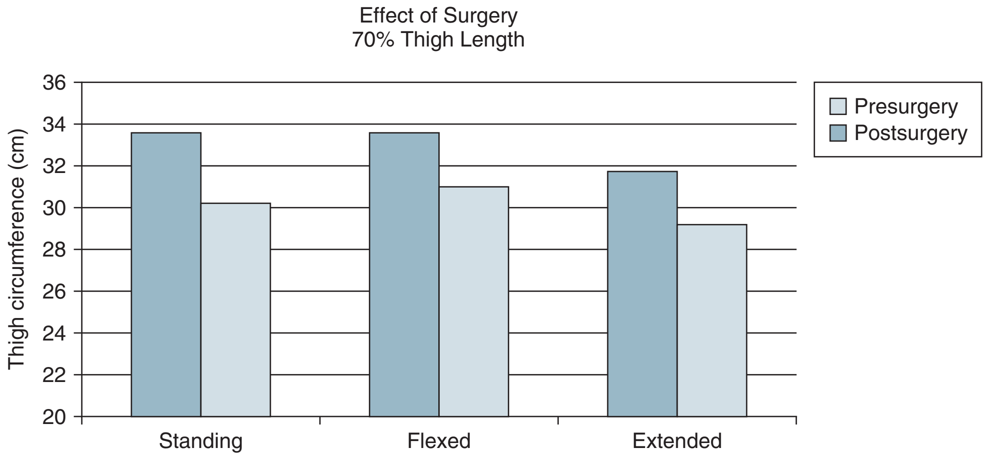


Figure 13-13 Thigh circumference measurements are sensitive enough to detect muscle atrophy 2 weeks after stifle surgery, indicating that this technique may be useful as a simple, cost-effective method of assessing limb mass.

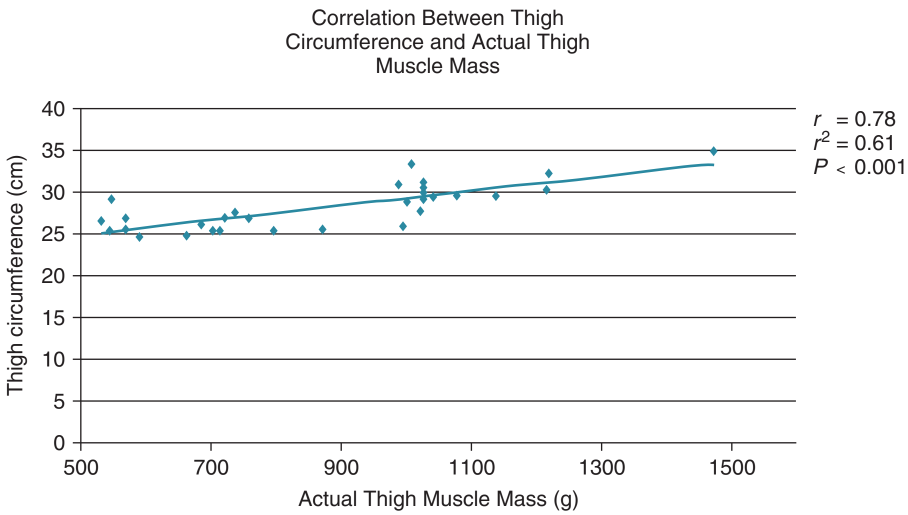


Figure 13-14 Thigh circumference has significant correlation with actual thigh muscle mass in dogs.

also considered and the volume of a cylinder is calculated. Although this method is simplistic because the thigh is not a true cylinder, the volume of a cylinder is easy to calculate and this estimate of muscle mass may be more clinically useful. Similar principles likely exist for circumference measurements of other limbs. Currently we prefer performing thigh circumference measurements with the hair short or clipped, the limb held in an extended position, at 70% of the length of the thigh, and with the animal relaxed, but not necessarily sedated. In the forelimb the length is measured from the proximal aspect of the olecranon to the distal aspect of the lateral styloid process; the limb circumference is measured at a point that is 20% of the limb length as measured from the proximal aspect of the lateral epicondyle of the humerus.

Limb circumference measures are commonly used in people to assess muscle atrophy, but relatively little is known regarding circumference measurement changes following knee injury and subsequent surgery. One study compared thigh and calf circumference measurements of affected and unaffected extremities before and after knee surgery for patients with acute and chronic knee injuries.[39] There were significant differences between affected and unaffected extremities at both the presurgery and postsurgery time periods of the acute and chronic groups. The bulk of the circumference measurement differences existed before surgery for both groups. In another study, the intrarater and inter-rater reliability of lower extremity circumference measurements in humans recovering from ACL reconstructive surgery demonstrated high correlation (0.82 to 1.0 and 0.72 to 0.97, respectively) for both the involved and uninvolved sides in several locations.[40]

Measurement of thigh circumference may be biased by the expectations of observers.[41] Use of anatomic landmarks and a tape measure with a spring tension device may help to minimize bias of the evaluator because the location of the measurement and the end tension of the tape can be standardized. A relatively small change in limb circumference may indicate much greater loss of muscle mass. A study of people with chronic unilateral patellofemoral pain indicated that although symptomatic limbs had significant reductions in limb circumference, very small reductions in thigh circumference (1%) were associated with significant reductions in muscle size (13%).[42]

The thickness of the skin and subcutaneous tissues may be another factor affecting thigh circumference measurements.[43] One investigation developed an equation using circumference and skinfold measurements for estimating anatomic cross-sectional area (CSA) of the quadriceps, hamstrings, and total thigh muscles.[44] Derived equations using midthigh circumference and anterior thigh skinfold assessment may be of value for estimating muscle CSA values when more sophisticated procedures are not available. Other studies have made similar assumptions that a cross-section of the thigh could be represented as a circle with concentric layers of skin, fat, muscle, and bone.[45] However, approximations of thigh muscle CSA may underestimate the fat-plus-skin compartment when calipers are used.

Although thigh circumference has been used as an effective method of measuring muscle size to evaluate the effect of an injury or effectiveness of an intervention, some studies in humans have shown this technique to be unreliable under certain conditions. Therefore other methods of measuring muscle size may be useful. Such modalities, although costly, include ultrasound scans, DEXA, CT, or MRI of muscles.

One study investigating the validity and reliability of measuring human quadriceps CSA with ultrasound scanning at the level of the mid-thigh found it to be a reliable method.[46] In a similar study, thigh circumference and segment volumes of the thigh were more strongly correlated to anterior versus posterior thigh musculature (when corrected for skinfold thickness) when B-mode ultrasound scanning was used.[43]

The question of whether various evaluation techniques are sensitive enough to detect changes in muscle mass was addressed in a training study of men.[47] Men were exposed to conditions designed to elicit differential hypertrophic adaptations following 21 sessions of squat training. A control group did no formal physical training. Tests used to evaluate muscle size included thigh circumference and quadriceps femoris and hamstring thicknesses via B-mode ultrasound. Thigh circumference and quadriceps femoris thickness were greater in groups that trained compared to than in controls.

Magnetic resonance imaging and computed tomography are promising reference methods for quantifying whole body and regional skeletal muscle mass. In people, arm and leg skeletal muscle CSA estimates obtained by standard MRI and CT methods had good correlation with corresponding cadaver values.[48] Magnetic resonance imaging used to estimate muscle CSA also had good correlation with actual cross-sections of anatomic specimens in other studies.[45,49] These findings strongly support the use of MRI and CT as methods to determine appendicular skeletal muscle mass in vivo.

Neurogenic muscle atrophy induced by crushing the sciatic nerve was examined using CT in dogs.[50] The CT number and CSA in denervated muscles decreased 1 to 2 weeks after denervation and were significantly less after 3 weeks. Examination with CT may be useful to evaluate neurogenic muscular atrophy.

Some work has compared various methods of muscle mass estimation in people. For example, the CSA and volume of the quadriceps femoris muscle using MRI was compared with B-mode ultrasound.[51] There was no significant difference in the CSA estimates or volume estimates when ultrasound and MRI were compared at several sites of the thigh.

Dual-energy x-ray absorptiometry is another noninvasive method of measuring lean tissue mass, but the estimations are generally limited to the whole body or an entire limb because it is difficult to determine lean tissue mass of relatively small regions. Although CT and MRI have been used to measure limb circumference and have the advantage that the area of large muscles may be measured, the equipment is expensive and interpretation must be performed by a person trained in these diagnostic modalities. Dual-energy x-ray absorptiometry is somewhat less expensive, and less training is required for interpretation of data, although patient positioning is critical to obtain accurate results.

The accuracy of DEXA for measuring total body fat–free mass and leg muscle mass at four leg regions was assessed in people.[52] Computed tomography of the legs was also performed. Fat-free mass and muscle mass by DEXA was positively associated with CT at all four leg regions ($r^2 = .86$ to .96), but the correlation of muscle mass with DEXA was higher than with CT in three regions. A comparison of DEXA with MRI to evaluate estimates of muscle and adipose tissue in human lower limb sections indicated high agreement between DEXA and MRI for muscle, but less so for adipose tissue.[53]

Muscle Strength

Little investigation has been performed in dogs regarding muscle strength estimates, likely because of the difficulty in making such measurements in animals. One study of Beagle dogs measured muscle strength as indicated by measuring maximum isometric extension torque of the hindlimb.[54] This required general anesthesia, a surgical approach to the thigh of the dog, placement of a stimulating electrode near the femoral nerve, and instrumentation of the limb to measure muscle torque while the muscle was electrically stimulated. Such invasive procedures are not clinically applicable. However, maximum isometric extension torque had strong correlation with muscle fiber diameter obtained from biopsies of the vastus lateralis muscle. Obtaining muscle biopsies is an invasive procedure, although less so than the technique for determining muscle torque, and less instrumentation is required.

Other studies of muscle strength of humans may provide information that may be applicable to dogs. One study determined the relationship between selected anthropometric dimensions and strength in resistance-trained athletes.[55] Athletes were measured following the completion of a 10-week resistance training program for one-repetition maximum lifts. The highest relationships existed between estimates of regional muscle mass (arm circumference, arm muscle CSA, and thigh circumference) and lifting performance. The fewer joints and muscle groups involved in a lift, the greater was the predictive accuracy from structural dimensions.

Although muscle mass is clearly associated with muscle strength, the type and strength of this relationship are less clear. The use of limb circumference measurements as an indication of muscle strength in humans is somewhat controversial. A predictive model was developed to evaluate the relationship of peak knee flexion and extension torque production to thigh circumference.[56] The authors concluded that peak knee torque production can be predicted with significant accuracy ($r^2 = .78$ to .87). A similar study assessed the correlation of maximal isometric force production of leg extensor muscles of elite human weight lifters with thigh circumference.[57] Maximal isometric force had significant correlation with thigh circumference, but surprisingly not with the mean muscle fiber area of the vastus lateralis muscle. The CSAs of fat, muscle, and bone tissues of limbs as well as maximal voluntary isokinetic strength were measured in men and women in another study.[58] Anatomic CSAs were determined by ultrasound of the upper arm and thigh. The isokinetic strength of the elbow and knee extensor and flexor muscles was measured using an isokinetic dynamometer. There was significant correlation between CSA and strength in all muscle groups except for the elbow extensors of the men and the elbow flexors of the women.

Another study correlated thigh circumference, muscle CSA by MRI, and isokinetic strength in patients 48 months after surgery for ACL injury.[59] There was a significant 1.8% decrease in thigh circumference, an 8.6% decrease in quadriceps CSA area by MRI, and a 10% decrease in average quadriceps torque in the involved extremities as compared with the uninvolved extremities. A positive correlation was found between MRI CSA and quadriceps and hamstring peak torque in involved and uninvolved extremities. A positive correlation among thigh circumference, quadriceps, and hamstring peak torque was found in uninvolved extremities but not in operated extremities. Another study assessed whether measurements of thigh circumference of patients undergoing unilateral meniscectomies and rehabilitation exercises would be an indicator of muscle power.[60] In this study muscle power was not predicted from thigh circumference measurements. Another study also found that the changes in muscle strength were not directly correlated to limb circumference. In this study isometric quadriceps strength and thigh circumference were determined in humans before and after a unilateral strength-training protocol of the quadriceps muscles for 5 weeks.[61] There were no significant changes in the untrained thighs. The trained quadriceps increased their isometric strength by 15%, whereas they changed their CSA by only 6%. Quadriceps hypertrophy was underestimated by measurements of thigh circumference. Nevertheless CSA measurements may indicate improvement in strength produced by training.

Although limb circumference may be associated with muscle power and strength in some situations, other factors may affect muscle strength besides muscle size. A single measurement of circumference in both thighs of subjects with unilateral injury may not be adequate to assess muscle function, but serial measurements over time may be of value as an index of muscle power. One study assessed the relationship between thigh circumference and muscle strength and power in noninjured and injured human subjects.[62] The correlation between the torque produced at the knee by the knee extensors and flexors and a single thigh circumference measurement was not significant. In contrast, repeated measurements over a 6- to 8-month period showed a significant relationship between change in thigh circumference and change in quadriceps power.

Muscle Injury

Evaluation of muscle injury during rehabilitation may provide important information, especially as it pertains to prevention of overuse injuries. Magnetic resonance spectroscopy (MRS) and MRI are powerful tools to study tissue biochemistry and provide precise anatomic visualization of soft tissue structures.[63] These techniques may be used to study exercise-induced muscle injury. Magnetic resonance spectroscopy measurements show an increase in the ratio of inorganic phosphate to phosphocreatine (Pi/PCr) after eccentric exercise. This increase could be due to either increases in extracellular Pi or resting muscle metabolism. Increased Pi/PCr is also seen during training programs and may indicate persistent muscle injury. Increased resting Pi/PCr with injury is not associated with altered metabolism during exercise. Elevations in resting Pi/PCr have been used to show increased susceptibility of dystrophic muscle to exercise-induced injury. Progressive clinical deterioration in dystrophic dogs is marked by impaired muscle metabolism, and the presence of low oxidative muscle fibers not seen in normal dogs. Unfortunately MRS is not commonly available for use in animals.

Magnetic resonance imaging shows changes following eccentric exercise that last up to 80 days after injury and can reflect muscle edema as well as longer lasting changes in the characteristics of cell water. Magnetic resonance imaging shows the precise localization of the injured area. Thus MRS can provide information on the metabolic response to injury, whereas MRI provides information regarding the site and extent of the injury. These tools are promising as aids in understanding exercise-induced muscle injury.

Body Composition

Assessment of body condition is important in rehabilitation patients because obesity has been associated with the exacerbation of some conditions, such as osteoarthritis. In addition many companion animals are obese or overweight, and this may affect performance and health. Conversely poor muscle mass may indicate inadequate nutrition, pathologic conditions, injuries, or poor conditioning. Morphometric techniques such as body condition scoring systems, commonly used in clinical practice, are simple, inexpensive, and generally reasonably accurate. Body conformation is assessed from the side and the dorsal aspect; the animal is also palpated over certain regions. These profiles are then compared with a standard and a body condition score is assigned (Figure 13-15). Other morphometric measures, including tape measurements of body areas, such as the cranial thoracic region and abdomen, are somewhat useful for assessing body composition, but because of the variation in body type in dogs, they may not be better than body condition scores.[64]

Although body condition scores are clinically useful, they are not as precise as other methods of noninvasive body composition assessments, such as DEXA analysis.[64] These units (Figure 13-16) measure lean body mass, fat, and bone mineral and are accurate within 2% of actual measurements. In addition many software programs allow the generation of body composition of various portions of the body, such as the left and right forelimbs and rear limbs. Although generally considered the gold standard for the noninvasive assessment of body composition, animals must be sedated or anesthetized to obtain information. Dual-energy x-ray absorptiometry units are relatively expensive and are limited to institutions at this time.

Skin caliper measurements, although having relatively good correlation to body composition in humans, are generally not very useful in dogs, perhaps because of differences in subcutaneous tissues and fat distribution.[64] Other methods of assessing body composition, such as diagnostic ultrasound, electrical impedance, and isotope dilution using the deuterium oxide method, have insufficient investigation, require specialized equipment, or are limited to research institutions.

Pain Assessment

Assessment of pain and discomfort is important in the physical rehabilitation of animals. Excessive discomfort may prevent or slow progress during treatment, but objective measurement of pain is difficult in animals because they do not verbalize the level of pain that they may be experiencing. Therefore pain assessment scores or questionnaires are generally used to evaluate behaviors that are believed to be associated with pain and discomfort (Box 13-2). In addition, ordinal or VASs are sometimes used in which owners are asked how painful they believe their pet is. Some physiologic parameters, such as heart rate, respiratory rate, and blood pressure, have been used to evaluate pain in the acute postoperative period, but these are generally not as useful for assessing pain in chronic conditions.

Multiple studies have examined the validity and clinical usefulness of pain scales and pain questionnaires. The

Nestlé PURINA

BODY CONDITION SYSTEM

TOO THIN

1 Ribs, lumbar vertebrae, pelvic bones and all bony prominences evident from a distance. No discernible body fat. Obvious loss of muscle mass.

2 Ribs, lumbar vertebrae and pelvic bones easily visible. No palpable fat. Some evidence of other bony prominence. Minimal loss of muscle mass.

3 Ribs easily palpated and may be visible with no palpable fat. Tops of lumbar vertebrae visible. Pelvic bones becoming prominent. Obvious waist and abdominal tuck.

IDEAL

4 **Ribs easily palpable, with minimal fat covering. Waist easily noted, viewed from above. Abdominal tuck evident.**

5 **Ribs palpable without excess fat covering. Waist observed behind ribs when viewed from above. Abdomen tucked up when viewed from side.**

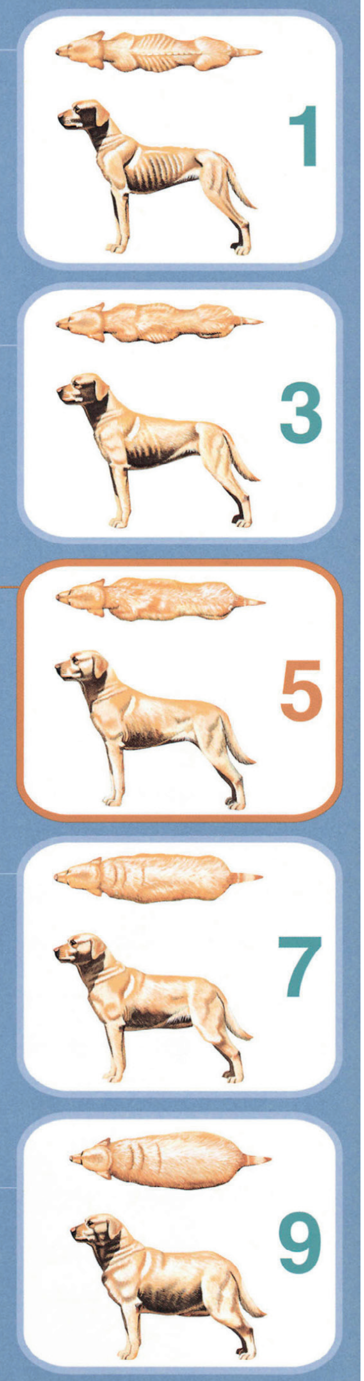
TOO HEAVY

6 Ribs palpable with slight excess fat covering. Waist is discernible viewed from above but is not prominent. Abdominal tuck apparent.

7 Ribs palpable with difficulty; heavy fat cover. Noticeable fat deposits over lumbar area and base of tail. Waist absent or barely visible. Abdominal tuck may be present.

8 Ribs not palpable under very heavy fat cover, or palpable only with significant pressure. Heavy fat deposits over lumbar area and base of tail. Waist absent. No abdominal tuck. Obvious abdominal distention may be present.

9 Massive fat deposits over thorax, spine and base of tail. Waist and abdominal tuck absent. Fat deposits on neck and limbs. Obvious abdominal distention.



The **BODY CONDITION SYSTEM** was developed at the Nestlé Purina PetCare Center and has been validated as documented in the following publications:

Mawby D, Bartges JW, Moyers T, et. al. ***Comparison of body fat estimates by dual-energy x-ray absorptiometry and deuterium oxide dilution in client owned dogs.*** *Compendium 2001; 23 (9A): 70*

Laflamme DP. ***Development and Validation of a Body Condition Score System for Dogs.*** *Canine Practice July/August 1997; 22:10-15*

Kealy, et. al. ***Effects of Diet Restriction on Life Span and Age-Related Changes in Dogs.*** *JAVMA 2002; 220:1315-1320*

Call 1-800-222-VETS (8387), weekdays, 8:00 a.m. to 4:30 p.m. CT

Nestlé PURINA

Figure 13-15 Body condition scores may be assigned to patients. The patient is assessed from the side and the dorsal aspect and the conformation is compared to a standard. *(Body condition score courtesy Iams Co., Dayton, OH.)*

Figure 13-16 A dual-energy x-ray absorptiometer may be used to measure lean body mass, fat mass, and bone mineral content.

Box 13-2 Simple Pain Assessment Score

0 No signs of pain during palpation of affected joint
1 Signs of mild pain during palpation of joint
2 Signs of moderate pain during palpation
3 Signs of severe pain during palpation
4 Dog will not allow examiner to palpate joint

canine brief pain inventory (CBPI) is based on the human Brief Pain Inventory (BPI) and has been shown to be reliable and valid in dogs with osteoarthritis,[12] and in a separate study on osteoarthritic dogs was shown to be sensitive enough to detect changes in pain with nonsteroidal antiinflammatory (NSAID) use.[13] The CBPI has also been validated for use in dogs with bone cancer.[65]

The Helsinki chronic pain index (HCPI) was created to assess chronic pain in dogs[11] (Table 13-1) and was found to be a valid and reliable tool for assessment of chronic pain by owners of dogs with osteoarthritis.[66] However, caution should be used with any subjective evaluation of pain in dogs with osteoarthritis. One study compared subjective evaluation of dogs receiving a placebo for the treatment of osteoarthritis and found that caregivers indicated that 40% of patients improved when evaluated by subjective measures, while measurement of ground reaction forces indicated no improvement.[67]

Other studies have evaluated the usefulness of questionnaires to evaluate acute pain in dogs. A short form of the Glasgow Composite Measure Pain Scale was able to identify those patients that required analgesic treatment from those that did not.[68] This pain scale evaluates six behavioral categories including vocalization, attention to wound, mobility, response to touch, demeanor, and posture/activity. Descriptions of the severity of each category are provided, and the items are placed in increasing order of pain intensity and numbered. The observer chooses the item within each category that best describes the dog's behavior and the scores are summed. The University of Melbourne Pain Scale incorporates heart rate, respiratory rate, and behavioral responses, such as response to palpation, activity, mental status, posture and vocalization. This scale is most useful in acute postoperative patients.[69]

Painful behavior in dogs may be indicated by whining, crying, or other forms of vocalizing when the animal moves or the affected body part is manipulated. Holding an affected limb in a tightly flexed or guarded position is frequently noticed after surgery. Animals may be particularly resentful of palpation and manipulation of the area. Painful behavior may also be exemplified by lying quietly in the back of a run or in the corner of a room with little desire to rise and move about. This may be a protective mechanism to avoid further injury and pain from moving about. The degree of pain in these patients is frequently underestimated because of their quiet nature.

Impressions of Owners and Veterinarians

Dogs and cats have different personalities and may respond very differently to various therapies. It is important for individuals who best know patients to evaluate their progress. For example, owners are often very aware of subtle changes in their pet's behavior. Although it is somewhat difficult to quantify these changes, these subjective findings often provide the therapist with important information regarding patient progress. If possible, the changes should be as objective as possible, such as recording the amount of time a pet spends playing rather than resting, measuring the distance that an animal is able to walk before needing to rest, or evaluating the length of time a neurologic patient is able to stand unassisted.

Owners should keep a log of daily activities to further define the progress of the patient at home. This may also provide incentive for owners to continue a home rehabilitation program if they can see the progress that is made from week to week and month to month. Poor compliance is a common reason for failure of a patient to progress. Regularly checking the log gives the veterinarian and owner an opportunity to evaluate the patient's progress, as well as make alterations to the rehabilitation program that account for an owner's ability to provide continued home care. In addition, videotaping the patient's activity on a regular schedule allows comparisons to be made over relatively long periods of time.

Return to Function

Return to function is perhaps the best indication of a successful outcome following treatment of an acute or chronic condition. The desired level of function depends on the intended use of the patient, which varies from a highly

Table 13-1 Pain Assessment in Dogs with Chronic Osteoarthritis

	SCORES FOR DOGS WITH CHD		SCORES FOR CONTROL DOGS		
Question Topic	**Median**	**Range**	**Median**	**Range**	***P* Value**
POSITIVE BEHAVIOR					
Appetite	0	0-3	0	0-2	0.59
Mood*	1	0-3	0	0-1	<0.001
Frequency of contact with human family members	1	0-2	0	0-2	0.086
Frequency of tail wagging	1	0-3	0	0-2	0.013
Activity	1.5	0-4	1	0-3	0.052
Play and games*	1	0-4	0	0-1	<0.001
NEGATIVE BEHAVIOR					
Excessive panting	1	0-4	0	0-2	<0.001
Licking of lips	0	0-4	0	0-3	0.97
Vocalization (audible complaining)*	1	0-3	0	0-1	<0.001
Vocalization when stretching hind legs caudally	2	0-4	0	0-1	<0.001
Aggressiveness toward humans	0	0-3	0	0-2	0.88
Aggressiveness toward other dogs	2	0-3	1	0-3	0.47
Aggressiveness toward dogs in its own pack	1	0-4	1	0-2	0.09
Submissiveness in the pack	1.5	0-4	2	0-4	0.27
LOCOMOTION					
Walking*	1	0-3	0	0-1	<0.001
Trotting*	1.5	0-4	0	0-1	<0.001
Pacing	1	0-4	2	0-4	<0.001
Galloping*	1	0-4	0	0-1	<0.001
Jumping*	2	0-4	0	0-1	<0.001
Climbing stairs	2	1-4	0	0-1	<0.001
Descending stairs	2	0-4	0	0-2	<0.001
Laying down*	2	0-4	0	0-1	<0.001
Getting up*	2.5	0-4	0	0-1	<0.001
Difficulty moving after rest*	2	0-4	0	0	<0.001
Difficulty moving after major activity*	3	1-4	0	0-1	<0.001
Chronic pain index (sum of answers to 11 questions)	19	7-35	2	0-5	<0.001

*Question selected for inclusion in chronic pain index. Values of $P < 0.005$ were considered significant.

Comparison of 25 questionnaire answers (scores, 0-4) provided by owners of 41 dogs with pain associated with canine hip dysplasia (CHD) with those provided by owners of 24 dogs with no pain.

From Hielm-Björkman AK, Kuusela E, Liman A, et al: Evaluation of methods for assessment of pain associated with chronic osteoarthritis in dogs. *J Am Vet Med Assoc* 222:1552-1558, 2003.

trained working dog or racing greyhound to a house pet. Some outcomes are relatively easy to quantify, such as winning a race in the same class that the animal raced in before an injury or being able to perform a particular task in the case of a working dog. Others are more nebulous, such as a return to acceptable house pet function. Regardless of the desired outcome, reasonable expectations must be set based on the severity of the condition being treated, and this must be conveyed to the owner to avoid future disappointment when a higher level of function is expected than may be achieved.

For example, it is unreasonable to expect an elderly arthritic dog to return to competitive coursing, but it may be reasonable to expect the dog to be able to play ball for short periods of time. In addition, expectations may need to be altered if the response to treatment is better or worse than expected. In severely affected neurologic patients, reasonable expectations should center on functional activities of daily living, such as being able to climb stairs to get in and out of the house, go outside to eliminate with minimal or no assistance, and take short walks with the owner.

Functional scales are used extensively in human medicine and rehabilitation in orthopedics, neurology, cardiopulmonary, pain, child development, and other areas. Functional scoring systems for specific injuries may be useful, such as a functional stifle score. Such scoring systems have been used for outcome assessment in people recovering from ACL surgery because an ACL may result in functional disability. Two of the most commonly used measures are the Lysholm Knee Scale and the Cincinnati Knee Rating Scale.[70,71] Although these scales are well validated in humans, the optimal method to measure the degree of functional disability is unknown in dogs. A useful system in dogs likely should incorporate limb use, lameness, stance, pain, muscle atrophy, range of motion, joint effusion, drawer motion, and functional activities to evaluate limb function following cranial cruciate ligament surgery (Table 13-2). A subjective and functional rating system in people related six activity levels to pain, swelling, giving-way, and overall activity.[72] In this study, activity level, symptoms, clinical laxity, meniscal damage, lower limb alignment, tibiofemoral crepitus, patellofemoral factors, rehabilitation, and patient compliance were identified as risk factors for future joint arthrosis.

Other functional tests that have been used in humans may be applied to dogs, with modification. For example, one study of people assessed the sensitivity of four different types of one-legged hop tests to determine alterations in lower limb function in ACL-deficient knees.[73] Comparisons were made between limb symmetry as measured by the hop tests and muscle strength, symptoms, and self-assessed function. Statistical trends were noted between abnormal limb symmetry on the hop tests and low-velocity quadriceps isokinetic test results. The time and distance that a dog will walk on the rear limbs with the forelimbs elevated (dancing) may provide similar information. Several easy to administer functional scoring scales have been developed for dogs with spinal cord injuries including one for acute injuries[74] (Box 13-3), and another that is based on the normal neurologic examination[75] (Box 13-4). An example is Figure 13-17 where dogs with intact deep pan recovered faster and to a greater degree than dogs without deep pain.

Box 13-3 Functional Scoring System in Dogs with Acute Spinal Cord Injuries

The five stages of recovery of use of pelvic limbs in dogs with spinal cord injuries. Each stage is subdivided on the basis of recovery patterns (i.e., yielding a scale of 0 to 14). Dogs were considered as weight-bearing when the full weight was born with joints extended for at least two steps (i.e., standing alone was not considered weight-bearing).

Stage 1

0 No pelvic limb movement and no deep pain sensation.
1 No pelvic limb movement with deep pain sensation.
2 No pelvic limb movement but voluntary tail movement.

Stage 2

3 Minimal non-weight-bearing protraction of the pelvic limb (movement of 1 joint).
4 Non–weight-bearing protraction of the pelvic limb with >1 joint involved <50% of the time.
5 Non–weight-bearing protraction of the pelvic limb with >1 joint involved >50% of the time.

Stage 3

6 Weight-bearing protraction of pelvic limb <10% of the time.
7 Weight-bearing protraction of pelvic limb 10% to 50% of the time.
8 Weight-bearing protraction of pelvic limb >50% of the time.

Stage 4

9 Weight-bearing protraction 100% of the time with reduced strength of pelvic limb. Mistakes >90% of the time (e.g., crossing of pelvic limbs, scuffing foot on protraction, standing on dorsum of foot, falling).
10 Weight-bearing protraction of pelvic limb 100% of the time with reduced strength. Mistakes 50% to 90% of the time.
11 Weight-bearing protraction of pelvic limb 100% of the time with reduced strength. Mistakes <50% of the time.

Stage 5

12 Ataxic pelvic limb gait with normal strength, but mistakes >50% of the time (e.g., lack of coordination with thoracic limb, crossing of pelvic limbs, skipping steps, bunny-hopping, scuffing foot on protraction).
13 Ataxic pelvic limb gait with normal strength, but mistakes made <50% of the time.
14 Normal pelvic limb gait.

From Olby NJ, De Risio L, Muñana KR, et al.: Functional scoring system in dogs with acute spinal cord injuries. *Am J Vet Res* 62:1628, 2001.

Quality of Life

Questionnaires to assess the quality of life of dogs have been an area of recent study. Dogs with spinal cord injuries have been studied and a questionnaire was developed and evaluated for owner-perceived quality of life. Results of this study[76] suggest that this questionnaire can be used to obtain owner-perceived quality of life assessments. A follow-up study[77] evaluated owner-perceived quality of life in 100 dogs with spinal cord injuries. An interesting finding was that ambulation was the factor that owners perceived as having the biggest impact on quality of life.

Health-related quality of life in dogs with cardiac disease has also been investigated.[78] Three hundred and sixty dogs were evaluated by veterinarians and scored in terms of clinical signs of cardiac disease. Owner questionnaires were significantly correlated with the FETCH (Functional Evaluation of Cardiac Health) questionnaire and with the ISACHC (International Small Animal Cardiac Health Council) classification. Results of this study suggest that the questionnaire is a valid and reliable method for assessing quality of life in dogs with cardiac disease. Owner questionnaires have also been studied to help

Table 13-2 Proposed Stifle Function Scale

Category	Section	Findings	Maximal Score	Score
Limb use at a Trot	Normal	No lameness and weight bearing on all strides	10	10
	Mild disuse	Lame but weight bearing on >95% of strides		6
	Moderate disuse	Lame but weight bearing on >50% and <95% of strides		4
	Severe disuse	Lame but weight bearing on <50% and >5% of strides		2
	Very severe disuse	Continuous non–weight-bearing lameness or weight bearing on <5% of strides		0
Limb use at a Walk	Normal	No lameness and weight bearing on all strides	10	10
	Mild disuse	Lame but weight bearing on >95% of strides		6
	Moderate disuse	Lame but weight bearing on >50% and <95% of strides		4
	Severe disuse	Lame but weight bearing on <50% and >5% of strides		2
	Very severe disuse	Continuous non–weight-bearing lameness or weight bearing on <5% of strides		0
Lameness at a trot	None	Normal locomotion	10	10
	Slight	Trots with a slight (barely perceptible) lameness, but strides appear to have normal length		8
	Mild	Trots with a mild lameness, but strides appear to have normal length		6
	Moderate	Trot with a moderate (obvious) lameness or a shortened stride length on affected side when trotting, but is bearing weight on that limb		4
	Severe	Is intermittently non–weight-bearing on that limb when trotting		2
	Very severe	Is completely non–weight-bearing on that limb when trotting		0
Lameness at a walk	None	Normal locomotion	10	10
	Slight	Walks with a slight (barely perceptible) lameness, but strides appear to have normal length		8
	Mild	Walks with a mild lameness, but strides appear to have normal length		6
	Moderate	Walks with a moderate (obvious) lameness or shortened stride length on affected side, but is bearing weight on that limb		4
	Severe	Is intermittently non–weight-bearing on that limb when walking		2
	Very severe	Is completely non–weight-bearing on that limb when walking		0
Stance	Normal	Stands with equal weight on both pelvic limbs	10	10
	Mild asymmetry	Bears less weight on the affected pelvic limb or limb trembles when standing		6
	Moderate asymmetry	Puts limb down for balance but bears weight <10% of normal weight		3
	Severe asymmetry	Does not bear weight on affected limb while standing		0

Continued

Table 13-2 Proposed Stifle Function Scale—cont'd

Category	Section	Findings	Maximal Score	Score
Stair climbing	Normal	No difficulty	5	5
	Mild	Has slight difficulty climbing steps		3
	Moderate	Skips steps or bunny hops		1
	Severe	Cannot climb stairs		0
Sitting and standing	Normal	Easily goes from a sitting to standing or a standing to sitting position. Sits and rises squarely.	5	5
	Mild	Sits or stands with some difficulty (slight hesitation or delay)		4
	Moderate	Sits or stands with difficulty (hesitation or delay)		2
	Severe	Cannot sit or stand without assistance		0
Dancing	Normal	Moves freely forward and backward	5	5
	Mild anomaly	Resists moving forward and backward		3
	Severe anomaly	Unable to bear weight on pelvic limbs during forward and backward dancing motion		0
Pain response (palpation)	None	No pain response is elicited during palpation of the joint	5	5
	Mild	Mild pain response (i.e., head turning) is elicited during palpation of the joint		3
	Moderate	Moderate pain response (i.e., slight vocalization, increased reaction) is elicited during palpation of the joint		2
	Severe	Severe pain response (i.e., immediate reaction, loud vocalization, attempt to bite) is elicited during palpation of the joint		0
Stifle Effusion	None	No effusion of stifle	5	5
	Mild	Slight loss of patella ligament distinctness		3
	Moderate	Patella ligament not distinct		2
	Severe	Cannot distinguish patella ligament due to effusion		1
Muscle atrophy	None	Normal muscle mass	10	10
	Mild	If disease is unilateral, thigh girth is 0% to 5% smaller than the opposite limb. Otherwise, a slight atrophy of the biceps femoris or other thigh muscles is noted/palpated.		6
	Moderate	If disease is unilateral, thigh girth is 6% to 10% smaller than the opposite limb. Otherwise, a moderate atrophy of the biceps femoris or other thigh muscles is noted/palpated.		4
	Severe	If disease is unilateral, thigh girth is >11% smaller than the opposite limb. Otherwise, a severe atrophy of the biceps femoris or other thigh muscles is noted/palpated.		0
Joint motion* (extension)	Normal or increased	Extension 160° or more	5	5
	Mild loss	Extension 150° to 159°		3
	Moderate loss	Extension 140° to 149°		1
	Severe loss	Extension <139°		0

Table 13-2 Proposed Stifle Function Scale—cont'd

Category	Section	Findings	Maximal Score	Score
Joint motion* (flexion)	Normal	Flexion 45° or less	5	5
	Mild loss	Flexion 46° to 50°		3
	Moderate loss	Flexion 51° to 60°		1
	Severe loss of flexion	Flexion >60°		0
Drawer Motion or Cranial Tibial Thrust (for TPLO or TTA)	None	Less than 2 mm	5	5
	Mild	2-4 mm		3
	Moderate	5-7 mm		1
	Severe	.7 mm		0
Total score				100

*Jaegger G, Marcellin-Little DJ, Levine D: Reliability of goniometry in Labrador retrievers. *Am J Vet Res* 63:979-986, 2002.

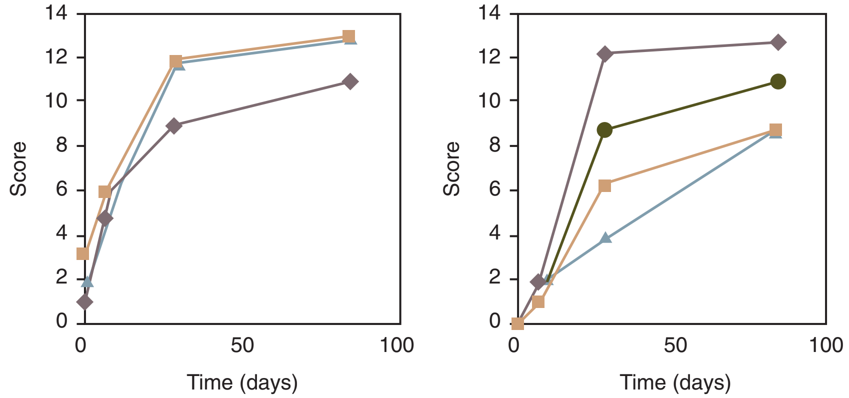


Figure 13-17 Typical course of recovery of pelvic limb function during a 3-month period in five dogs with spinal cord injuries that were nonambulatory paraparetics or paraplegics with intact deep pain sensation (left) and without deep pain sensation (right). Recovery scores are based on the system described in Box 13-3. (A score of 0 is poor, with no limb movement or deep pain; a score of 14 is best, with recovery of normal pelvic gait.) (From Olby NJ, De Risio L, Muñana KR, et al.: Functional scoring system in dogs with acute spinal cord injuries. *Am J Vet Res* 62:1628, 2001.)

Box 13-4 Texas Spinal Cord Injury Score for Dogs

Gait[a]

0 No voluntary movement seen when supported
1 Intact limb protraction with no ground clearance
2 Intact limb protraction with inconsistent ground clearance
3 Intact limb protraction with consistent ground clearance (>75%)
4 Ambulatory, consistent ground clearance with moderate paresis-ataxia (will fall occasionally)
5 Ambulatory, consistent ground clearance with mild paresis-ataxia (does not fall, even on slick surfaces)
6 Normal gait

Proprioceptive Positioning[b]

0 Absent response
1 Delayed response
2 Normal response

Nociception[c]

0 No deep nociception
1 Intact deep nociception, no superficial nociception
2 Nociception present

[a]Ground clearance refers to the ability to lift the limb off of the ground when it is being protracted.
[b]Proprioceptive positioning is performed by supporting the dog's weight and gently placing the dorsum of the paw on the ground. A delayed response is indicated by a >2-second lag between paw placement and correction.
[c]Deep nociception is measured by cross-clamping the distal limb or nail bed with hemostats. Superficial nociception is tested by pinching the interdigital webbing with hemostats.
From Levine GJ, Levine JM, Budke CM, et al: Description and repeatability of a newly developed spinal cord injury scale for dogs. *Prev Vet Med* 89:121-127, 2009.